CURRICULUM VITAE

Marjan Rashidi

CORRESPONDENCE

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EDUCATION

University of California, Irvine
Doctor of Science, Ongoing
- Major: Cognitive Science
  GPA: 4.00

University of California, Irvine
Master of Science, 2023
- Major Cognitive Neuroscience
  GPA:3.95

Iran University of Science and Technology
Master of Science, 2019
- Major: Architectural Engineering
  GPA: 3.61

Mazandaran University
Bachelor of Science, 2014
- Major: Architectural Engineering

Pre-University School
Pre-University Certificate and High School Diploma, 2010
- Major: Mathematics and Physics
  GPA: 4.00

RELEVANT COURSEWORK

Machine Learning, Computational Neuroscience, Reinforcement Learning, Intro to Programming (MATLAB), Computational Lab Skills (MATLAB), Intro to Database Management (SQL), Statistical Models (1&2), Bayesian Statistics, Cortical Neuroscience

RESEARCH EXPERIENCE

Deep Learning for Brain–Behavior Prediction Using Structural MRI Nov 2024 – Present

- Used Convolutional Neural Networks (CNNs) and Vision transformers (ViTs/Attention models) to explore how T1-weighted MRI scans can be used to predict performance in small-scale spatial tasks such as Mental Rotation and Paper Folding.

Alzheimer's Disease (AD) Project

Structural Brain Changes in AD Sep. 2024 – Nov 2024

- Analyzed white matter tracts of Alzheimer's Disease (AD) patients using the ADNI dataset, employing fiber tracking techniques to compare them with control and Mild Cognitive Impairment (MCI) groups.
- Applied statistical methods to assess differences between groups, contributing to a better understanding of structural brain changes associated with AD.

IndivRobotics Project

Brain Structure and Spatial Navigation Aug. 2021 – Sep 2024

- Designed a desktop virtual reality navigation task in Unity.
- Managed participant recruitment and conducted experiments in an immersive virtual reality lab (CAVERN), ensuring adherence to protocols and participant safety.
- Assisted in the collection and analysis of MRI data.
- Analyzed brain structures (including grey matter volume in several brain regions, cortical thickness, and white matter microstructures) using advanced neuroimaging techniques such as Advanced Normalization Tools (ANTs) and Correlational Tractography.
- Investigated the relationships between brain morphology and cognitive functions, with a particular focus on navigational and spatial abilities.
- Investigated the effects of stress and anxiety on human navigation abilities using statistical methods.
- Developed a *neural network* to predict human spatial abilities based on brain structural data, utilizing brain decoding methods to uncover patterns that link brain structure to cognitive performance using *PyTorch* in Python.

Neural Network Project Aug. 2024

Predicting Trial Types from BOLD Signals

- Developed a Multi-Layer Perceptron (MLP) model to predict trial types (fearful vs. neutral) based on fMRI BOLD signals from the Human Connectome Project (HCP) dataset.
- Implemented a backward elimination technique for feature selection and model training.

Reinforcement Learning Project

Model-based and Model-free comparisons

- Explored the performance of model-free and model-based reinforcement learning algorithms in a maze navigation task.
- Implemented *Q-learning* as a *Model-Free* algorithm and *Q planning* as a *Model-Based* counterpart using *Tensorflow*/*Keras* packages in Python.

Computational Neuroscience Project

Working Memory and Neural Mechanisms

- Conducted a study simulating working memory activity during a delay period using a Recurrent *Neural Network (RNN)* to analyze the persistent activity of prefrontal cortex (PFC) neurons.
- Investigated the effects of different stimulus orientations on neural firing rates in the context of direction selectivity.

WORK EXPERIENCE

Mentoring Dec 2024 – Present

Department of Neurobiology and Behavior

- Mentored undergraduate students on their Undergraduate Research Opportunities Program (UROP) projects, providing guidance and support throughout the research process.
- Conducted weekly meetings to monitor progress, address challenges, and foster students' academic and professional growth.

Teaching Assistant Sep 2021– April 2023

Department of Cognitive Sciences

- Led discussions and facilitated engaging classroom activities for students in the *Personality Theory* course.
- Provided academic support and guidance to students, helping them to understand key concepts and improve their performance.
- Led discussions in *Psych Fundamentals* course.
- Led discussions in *People in Society* course.

ACADEMIC PAPERS

Journal Papers:

1. Rashidi, M., Hegarty, M., Chrastil, ER. (2024). Cortical thickness and gray matter volume supporting object-based and egocentric perspective transformations (preregistered on OSF)

2. Rashidi, M., He, C., Hegarty, M., Chrastil, ER. (2024). The role of chronic stress in spatial navigation in humans (preregistered on OSF)

3. Rashidi, M., Hegarty, M., Chrastil, ER. (2024). Spatial navigation, sex differences, and stress (in prep)

4. Munns, M. E., Tu, A. S., He, C., Rashidi, M., Zhou, M., Hatamian, N., Starrett, M. J., Stark, C. E. L., Krichmar, J. L., Chrastil, E. R., & Hegarty, M. (2024) Individual differences in human navigation ability: Representations, predictors, and strategies (preregistered on OSF)

5. Nelson, M., Munns, ME., He, C., Rashidi, M, Chrastil, ER, and Hegarty, M. (2024) Everyday Spatial Experience, Video Games, and Their Influence on Spatial Abilities (preregistered on OSF)

6. Bain, R., Rashidi, M., Chrastil, ER. (2026) Pairwise Learning Increases Data Efficiency in Deep Neural Networks (Under review at IEEE Transactions on Pattern Analysis and Machine Intelligence)

7. Rashidi, M., Bain, R., Stark, C. E. L., Krichmar, J. L., Hegarty, M., Chrastil, ER. (2026) Inferring continous cognitive traits from limited neuroimaging data using pairwise convolutional neural networks (in prep)

8. Derdeyn, P., Chen, L., Chen, H., et al. (2026). Leveraging endogenous resilience mechanisms in retrosplenial cortex to slow progression of Alzheimer's disease (Under review at Neuron)

POSTER PRESENTATIONS

Society for Neuroscience (sfn) Nov.2025
San Diego, California

Psychonomic Society Nov. 2024
New York City, New York

Psychonomic Society Nov. 2023
San Francisco, California

International Conference on Learning and Memory (LEARNMEM) April. 2023
Huntington Beach, California

TECHNICAL SKILLS

Computer Languages

- Python, R, MATLAB, SQL

Data Science and Machine Learning

- PyTorch, Keras, TensorFlow, Scikit-Learn, Pandas, Numpy, Matplotlib, Computer vision

Other Software

- Unity, Microsoft Office, Revit Architecture, AutoCAD, Lumion, Sketchup, Photoshop, Illustrator

PORTFOLIO

https://marjanrsd.github.io/ Aug. 2021 – Present

- Web portfolio of personal, professional, and school projects. Links to github, LinkedIn, etc